

LAB EXPERIMENT RECORD

Set 4: Product Inventory Management

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To visualize product inventory data using a bar chart, category-wise bar visualization, and scatter plot, and to develop an interactive Tableau dashboard combining the inventory charts.

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Additional fields used in the submitted visualizations: Product Category and Product Price. The executed R scatter plot uses prices 500, 750, 400, 900, and 600 for Products A-E respectively. The Tableau category view uses Electronics, Furniture, and Stationery to distinguish products.

PROGRAM 1: Bar Chart of Product Inventory Quantity

Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Dataset

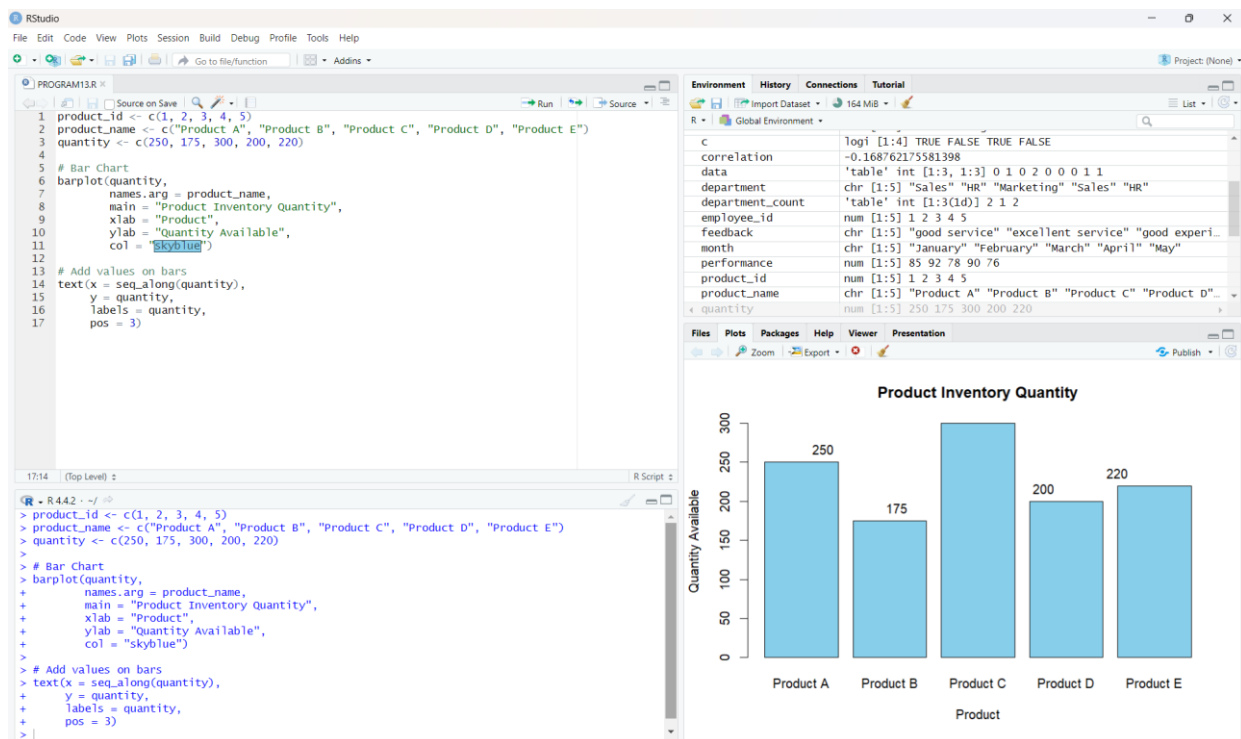
```
product_id <- c(1, 2, 3, 4, 5)
product_name <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)
```

Bar Chart

```
barplot(quantity,
        names.arg = product_name,
        main = "Product Inventory Quantity",
        xlab = "Product",
        ylab = "Quantity Available",
        col = "skyblue")
```

Add values on bars

```
text(x = seq_along(quantity),
     y = quantity,
     labels = quantity,
     pos = 3)
```



Output 1: Product Inventory Quantity Bar Chart

Observation: Product C has the highest available quantity of 300 units, while Product B has the lowest quantity of 175 units.

PROGRAM 2: Category-wise / Stacked Bar Visualization

Generate a category-wise bar visualization to show product quantities within different product categories. In the submitted Tableau view, Category is used on Color and Product Name is used on Columns.

Product category mapping used in the Tableau visualization

Product A -> Electronics

Product B -> Electronics

Product C -> Furniture

Product D -> Furniture

Product E -> Stationery

Tableau view

Columns: Product Name

Rows: SUM(Quantity Available)

Color: Category

Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
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4	Product D	200
5	Product E	220

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.
4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Output 2: Category-wise Inventory Visualization in Tableau

Observation: The visualization distinguishes the products by category. Product A and Product B are shown as Electronics, Product C and Product D as Furniture, and Product E as Stationery. Product C has the largest quantity among the displayed products.

Note: The submitted screenshot is a category-colored bar view rather than a single stacked bar per category. It records the Tableau visualization actually created for the experiment.

PROGRAM 3: Scatter Plot of Product Price vs Quantity Available

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
product_name <- c("Product A", "Product B", "Product C",
                 "Product D", "Product E")
```

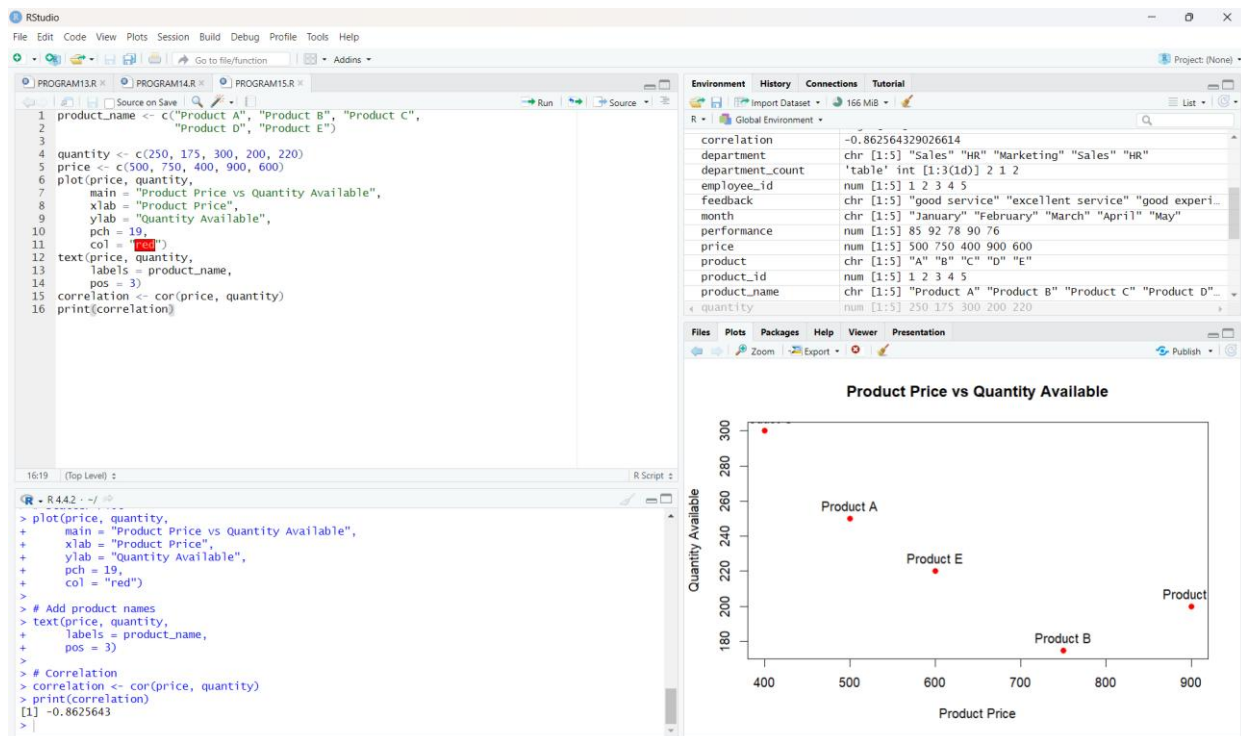
```
quantity <- c(250, 175, 300, 200, 220)
price <- c(500, 750, 400, 900, 600)
```

```
plot(price, quantity,
     main = "Product Price vs Quantity Available",
     xlab = "Product Price",
     ylab = "Quantity Available",
     pch = 19,
     col = "red")
```

```
text(price, quantity,
     labels = product_name,
     pos = 3)
```

```
# Correlation
```

```
correlation <- cor(price, quantity)
print(correlation)
```



Output 3: Product Price vs Quantity Available Scatter Plot

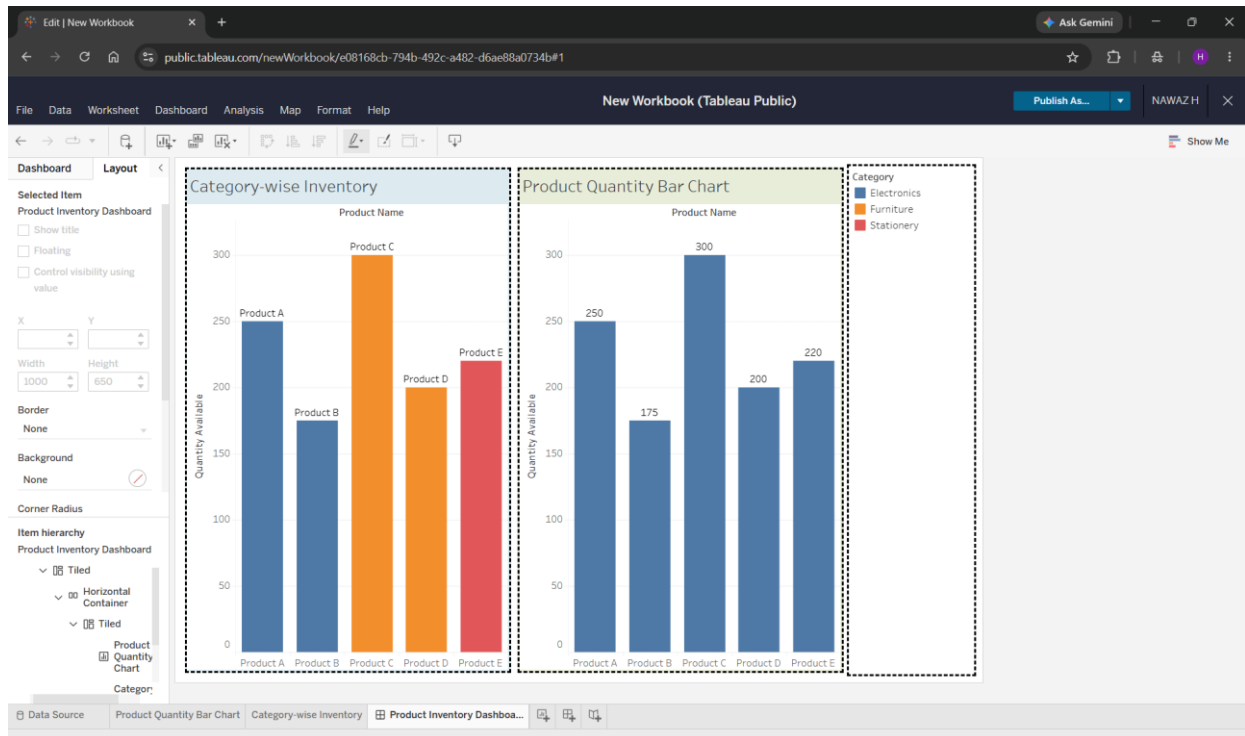
Finding: For the five products in the dataset, the plotted points show a generally negative relationship between price and quantity available. The calculated correlation is approximately -0.8626, indicating a strong negative linear association within these five observations.

PROGRAM 4: Interactive Tableau Dashboard

Develop a Tableau dashboard with the bar chart and category-wise bar chart to allow users to interact with the data.

Tableau Dashboard Steps

1. Connect the Product Inventory dataset in Tableau Public.
2. Create a Bar Chart using Product Name on Columns and SUM(Quantity Available) on Rows.
3. Create the Category-wise Inventory view using Product Name on Columns, SUM(Quantity Available) on Rows, and Category on Color.
4. Create a new dashboard named "Product Inventory Dashboard".
5. Place the two worksheets side by side in the dashboard.
6. Keep the Category legend visible so users can distinguish product categories.
7. Add suitable titles and labels for clear interpretation.



Output 4: Product Inventory Tableau Dashboard

Observation: The dashboard combines the Product Quantity Bar Chart and Category-wise Inventory visualization in one view. The Category legend allows the products to be distinguished by category.

Procedure

1. Create the Product Inventory dataset with Product ID, Product Name, and Quantity Available.
2. Use `barplot()` in R to visualize the quantity available for each product.
3. Use product category information in Tableau to distinguish inventory products by category.
4. Create the Product Price vs Quantity Available scatter plot in R and calculate the correlation using `cor()`.
5. Open Tableau Public and create the Product Quantity Bar Chart and Category-wise Inventory visualization.
6. Combine both visualizations in a Tableau dashboard named Product Inventory Dashboard.
7. Add titles, labels, category legend, and suitable formatting for interactive exploration.

Result

The Product Inventory dataset was successfully visualized using a bar chart, category-wise visualization, and scatter plot. The relationship between product price and quantity available was examined using a correlation value of approximately -0.8626. A Tableau dashboard was also created by combining the product quantity bar chart and category-wise inventory visualization.

Key R Functions / Tableau Tools Used

Function / Tool	Purpose
<code>barplot()</code>	Creates the product quantity bar chart.
<code>text()</code>	Displays quantity values above the bars and product labels on the scatter plot.
<code>plot()</code>	Creates the scatter plot of product price and quantity.
<code>cor()</code>	Calculates the correlation coefficient.
Tableau Columns / Rows	Places dimensions and measures to build the charts.
Tableau Color	Distinguishes products according to their category.
Tableau Dashboard	Combines the two inventory visualizations for interactive exploration.